COSC2637 Big Data Processing
Lab 4 - Tutorial/Lab class (week 5)

Task 1. Cooccurrence Matrix

Develop a python MapReduce program to construct cooccurrence matrix with Pairs approach and Strips approach respectively. The two approaches have been discussed in week 4 lecture in detail.

- **Pairs approach**

  The source codes have been provided on course week 5 module on Canvas. Following the instructions in previous labs, upload to AWS EMR cluster the following files

  mapper.py

  reducer.py

  run.sh

  trip.txt

  – run.sh must be runnable by running

  ```
  chmod +x run.sh
  ```

  – need copy trip.txt to HDFS, in my case, in folder

  ```
  hadoop fs -copyFromLocal trip.txt /
  ```

  Note the first two items in the output of mapper form the key. As instructed in previous lab, this is specified in run.sh

```
#!/bin/bash

# Remove the HDFS output directory forcefully and recursively.
hadoop fs -rm -r -f /output

hadoop jar /usr/lib/hadoop/hadoop-streaming.jar \
-D stream.map.output.field.separator="\t" \
-D stream.num.map.output.key.fields=2 \
-D mapreduce.job.reduces=3 \
-files ./mapper.py,./reducer.py \
-mapper ./mapper.py \
-reducer ./reducer.py \
-input /trip.txt \
-output /output
```

If running successfully, the output looks like:

```
[hadoop@ip-192-168-26-208 ~]$ hadoop fs -cat /output/part-00000
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/usr/lib/hadoop/lib/slf4j-log4j12-1.7.25.jar!/
org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/usr/share/aws/emr/emrfs/lib/slf4j-log4j12-1.7
.12.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.slf4j.impl.Log4jLoggerFactory]
I       am      4
I       girl    1
I       student 1
I       teacher 1
a       am      4
am      I       4
am      a       4
am      boy     1
am      girl    1
boy     I       1
boy     a       1
boy     am      1
student am      1
```

```
[hadoop@ip-192-168-26-208 ~]$ hadoop fs -cat /output/part-00001
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/usr/lib/hadoop/lib/slf4j-log4j12-1.7.25.jar!/
org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/usr/share/aws/emr/emrfs/lib/slf4j-log4j12-1.7
.12.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.slf4j.impl.Log4jLoggerFactory]
I       a       4
I       boy     1
a       I       4
a       student 1
a       teacher 1
am      student 1
girl    I       1
girl    a       1
student I       1
student a       1
teacher am      1
```

```
[hadoop@ip-192-168-26-208 ~]$ hadoop fs -cat /output/part-00002
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/usr/lib/hadoop/lib/slf4j-log4j12-1.7.25.jar!/
org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/usr/share/aws/emr/emrfs/lib/slf4j-log4j12-1.7
.12.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.slf4j.impl.Log4jLoggerFactory]
a       boy     1
a       girl    1
am      teacher 1
girl    am      1
teacher I       1
teacher a       1
```

- **Strips approach**

The source codes have been provided on course week 5 module on Canvas. Following the instruction in previous labs, upload to AWS EMR cluster the following files

mapper.py

reducer.py

run.sh

trip.txt

– run.sh must be runnable by running

```
chmod +x run.sh
```

– need copy trip.txt to HDFS, in my case, in folder

```
hadoop fs -copyFromLocal trip.txt /
```

Different from Pairs approach, the first item in the output of mapper is the key. This is default setting. No need to specify anything in run.sh

```
#!/bin/bash

# Remove the HDFS output directory forcefully and recursively.
hadoop fs -rm -r -f /output20

hadoop jar /usr/lib/hadoop/hadoop-streaming.jar \
-D mapreduce.job.reduces=2 \
-files ./mapper.py,./reducer.py \
-mapper ./mapper.py \
-reducer ./reducer.py \
-input /trip.txt \
-output /output20
```

If running successfully, the output looks like:

```
[hadoop@ip-192-168-26-208 ~]$ hadoop fs -cat /output20/part-00000
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/usr/lib/hadoop/lib/slf4j-log4j12-1.7.25.jar!/
org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/usr/share/aws/emr/emrfs/lib/slf4j-log4j12-1.7
.12.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.slf4j.impl.Log4jLoggerFactory]
I;am,4;a,4;boy,1;girl,1;student,1;teacher,1
a;I,4;am,4;teacher,1;boy,1;student,1;girl,1
student;I,1;am,1;a,1
```

```
[hadoop@ip-192-168-26-208 ~]$ hadoop fs -cat /output20/part-00001
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/usr/lib/hadoop/lib/slf4j-log4j12-1.7.25.jar!/
org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/usr/share/aws/emr/emrfs/lib/slf4j-log4j12-1.7
.12.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.slf4j.impl.Log4jLoggerFactory]
am;I,4;a,4;student,1;teacher,1;boy,1;girl,1
boy;I,1;am,1;a,1
girl;I,1;am,1;a,1
teacher;I,1;am,1;a,1
```

Task 2. Cooccurrence Relative Frequencies

Develop a python MapReduce program to construct cooccurrence relative frequencies with Pairs approach and Strips approach respectively. The two approaches have been discussed in week 4 lecture in detail.

- **Pairs approach**

  The source codes have been provided on course week 5 module on Canvas. Following the instruction in previous labs, upload to AWS EMR cluster the following files

  mapper.py

  reducer.py

  run.sh

  trip.txt

  – run.sh must be runnable by running

  ```
  chmod +x run.sh
  ```

  – need copy trip.txt to HDFS, in my case, in folder

  ```
  hadoop fs -copyFromLocal trip.txt /
  ```

  Note the first two items in the output of mapper form the key. But we must use the left word of key for partitioner, that is, all keys with the same left word must be shuffled to the same reducer in order to count the total number for the left word. As instructed in previous lab, this is specified in run.sh

```
#!/bin/bash

# Remove the HDFS output directory forcefully and recursively.
hadoop fs -rm -r -f /output

#-D stream.map.output.field.separator="\t" note if seperator is \t, do not specify this line !!
#-D map.output.key.field.separator="\t"   note if seperator is \t, do not specify this line !!
hadoop jar /usr/lib/hadoop/hadoop-streaming.jar \
-D stream.num.map.output.key.fields=2 \
-D mapred.text.key.partitioner.options=-k1,1 \
-D mapreduce.job.reduces=3 \
-files ./mapper.py,./reducer.py \
-mapper ./mapper.py \
-reducer ./reducer.py \
-input /trip.txt \
-output /output \
-partitioner org.apache.hadoop.mapred.lib.KeyFieldBasedPartitioner
```

If running successfully, the output looks like below. Note the total number of left word is shown in the 4th column. The relative frequency is the number in 3rd divided by the number in 4th column.

```
[hadoop@ip-192-168-26-208 ~]$ hadoop fs -cat /output/part-00000
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/usr/lib/hadoop/lib/slf4j-log4j12-1.7.25.jar!/
org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/usr/share/aws/emr/emrfs/lib/slf4j-log4j12-1.7
.12.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.slf4j.impl.Log4jLoggerFactory]
boy     I       1       3
boy     a       1       3
boy     am      1       3
```

```
[hadoop@ip-192-168-26-208 ~]$ hadoop fs -cat /output/part-00001
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/usr/lib/hadoop/lib/slf4j-log4j12-1.7.25.jar!/
org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/usr/share/aws/emr/emrfs/lib/slf4j-log4j12-1.7
.12.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.slf4j.impl.Log4jLoggerFactory]
I       a       4       12
I       am      4       12
I       boy     1       12
I       girl    1       12
I       student 1       12
I       teacher 1       12
a       I       4       12
a       am      4       12
a       boy     1       12
a       girl    1       12
a       student 1       12
a       teacher 1       12
girl    I       1       3
girl    a       1       3
girl    am      1       3
teacher I       1       3
teacher a       1       3
teacher am      1       3
```

```
[hadoop@ip-192-168-26-208 ~]$ hadoop fs -cat /output/part-00002
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/usr/lib/hadoop/lib/slf4j-log4j12-1.7.25.jar!/
org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/usr/share/aws/emr/emrfs/lib/slf4j-log4j12-1.7
.12.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.slf4j.impl.Log4jLoggerFactory]
am      I       4       12
am      a       4       12
am      boy     1       12
am      girl    1       12
am      student 1       12
am      teacher 1       12
student I       1       3
student a       1       3
student am      1       3
```

- **Strips approach**

  The source codes have been provided on course week 5 module on Canvas. Following the instruction in previous labs, upload to AWS EMR cluster the following files

  mapper.py

  reducer.py

  run.sh

  trip.txt

  – run.sh must be runnable by running

  ```
  chmod +x run.sh
  ```

  – need copy trip.txt to HDFS, in my case, in folder

  ```
  hadoop fs -copyFromLocal trip.txt /
  ```

  Different from Pairs approach, the first item in the output of mapper is the key. This is default setting. No need to specify anything in run.sh:

```
#!/bin/bash

# Remove the HDFS output directory forcefully and recursively.
hadoop fs -rm -r -f /output20

hadoop jar /usr/lib/hadoop/hadoop-streaming.jar \
-D mapreduce.job.reduces=2 \
-files ./mapper.py,./reducer.py \
-mapper ./mapper.py \
-reducer ./reducer.py \
-input /trip.txt \
-output /output20
```

  If running successfully, the output looks like below. Note the total number of left word is shown in the last column. The relative frequency is the number in the list divided by the number in the last column.

```
[hadoop@ip-192-168-26-208 ~]$ hadoop fs -cat /output20/part-00000
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/usr/lib/hadoop/lib/slf4j-log4j12-1.7.25.jar!/
org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/usr/share/aws/emr/emrfs/lib/slf4j-log4j12-1.7
.12.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.slf4j.impl.Log4jLoggerFactory]
I       am,4;a,4;boy,1;girl,1;student,1;teacher,1       total = 12
a       I,4;am,4;teacher,1;boy,1;student,1;girl,1       total = 12
student I,1;am,1;a,1    total = 3
```

```
[hadoop@ip-192-168-26-208 ~]$ hadoop fs -cat /output20/part-00001
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/usr/lib/hadoop/lib/slf4j-log4j12-1.7.25.jar!/
org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/usr/share/aws/emr/emrfs/lib/slf4j-log4j12-1.7
.12.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.slf4j.impl.Log4jLoggerFactory]
am      I,4;a,4;student,1;teacher,1;boy,1;girl,1        total = 12
boy     I,1;am,1;a,1    total = 3
girl    I,1;am,1;a,1    total = 3
teacher I,1;am,1;a,1    total = 3
```

Don’t forget to terminate your cluster at end of the lab class!

Also, if trying out any these exercises outside of the lab class, make sure you terminate your cluster and DO NOT leave clusters running idle for days.

```
./terminate_cluster.sh
```